Toronto Zoo Visitor Experience Insights

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Outline

1. Extracting top themes of visitor dissatisfaction (Topic Modeling)
2. Identifying pain points and hidden gems (Quadrant Mapping)
3. Pinpointing top areas of improvement (Key Driver Analysis)
4. Determining experience specific customer personas (Customer Segmentation)

Key Takeaways

- Across all analyses, customers reported being most frustrated with the core zoo experience
- In particular, the largest increases in zoo recommendations and visits could be gained from:
  - improving animal visibility, availability and variety
  - reducing the impact of construction closures and the overall admission fee
- Further, the best subareas to the zoo to target for such improvements would be:
  - The African Savannah
  - The Eurasian Wilds
  - The Tundra Trek
- However, these changes will be slow and costly. In the meantime, lower-hanging fruit changes can be made to fix zoo amenities.
- In this area, the largest increases in zoo recommendations and visits could be gained from improving:
  - Food services (The clear standout).
  - Operational Reliability (e.g. Zoomobile delays)

Extracting Top Themes of Customer Dissatisfaction

Complaint Themes:

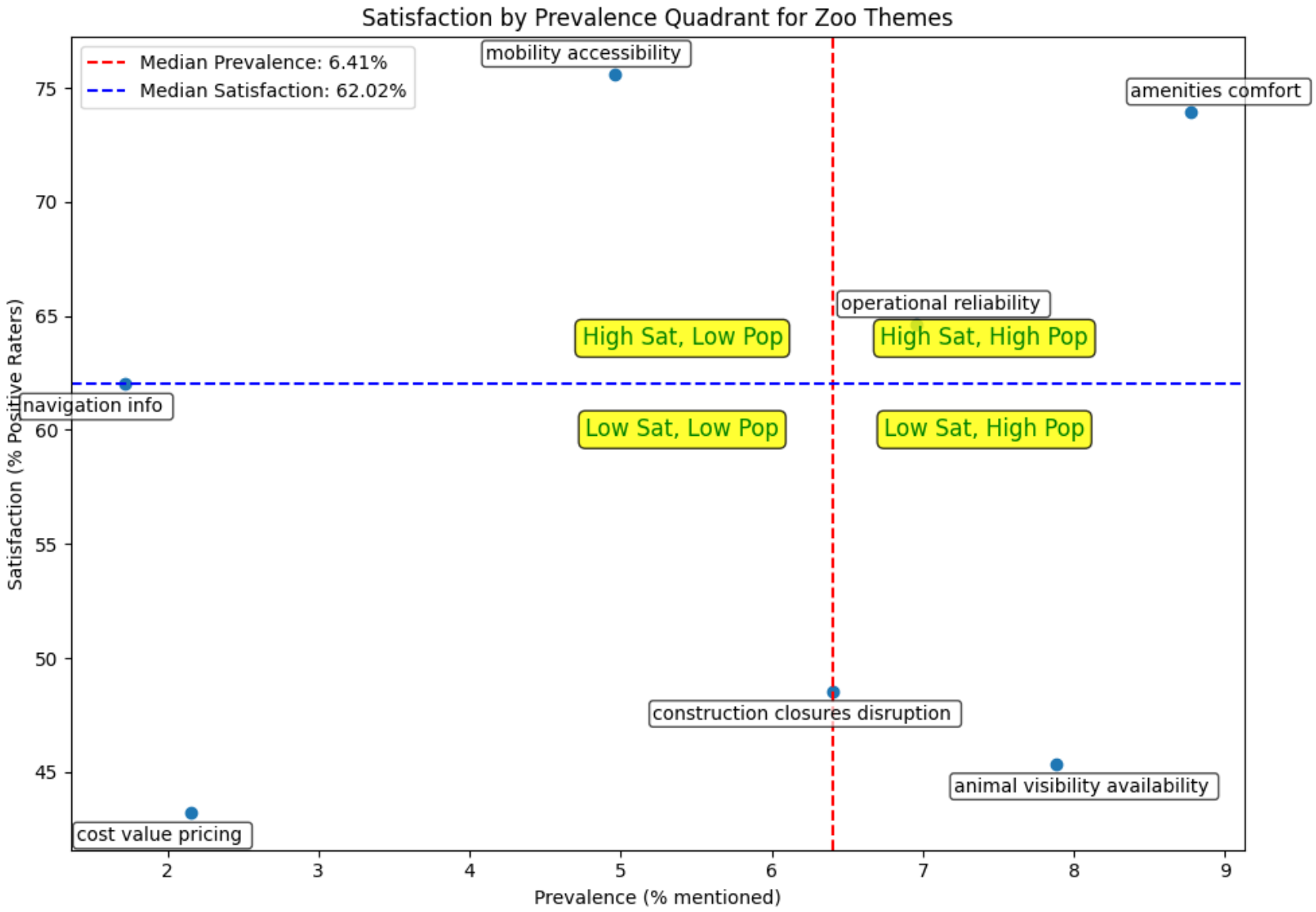
- Open feedback entries were cleaned and grouped into topics using topic modeling (BERTopic).
- High-friction topics were then determined using sentiment analysis.
- These high-friction topics were then grouped semantically into 7 themes of customer dissatisfaction.

Complaint Theme	Prominent Sub-Topics	% of Total Respondents
Mobility, Accessibility	Steep terrain, wheelchair and stroller access, narrow paths, lack of automatic doors	4.96%
Navigation, Information Access	Lack of signage indicating directions or closures, lack of website info on displays/closures	1.71%
Animal Availability, Visibility	Animals not available or visible in exhibits	7.88%
Construction, Closures, Disruption	Construction, subarea closure limiting access/experience	6.41%
Amenities, Comfort, Cleanliness	Washrooms messy or unavailable, food services expensive, slow, limited or unsatisfactory.	8.77%
Cost, Value, Pricing	Admission price too expensive, value for cost too high, parking too expensive	2.15%
Operational Reliability	Zoomobile wait times and breakdowns, parking and ticket payment issues.	6.95%

Identifying Popular Pain Points

Pain Points: Complaint Themes

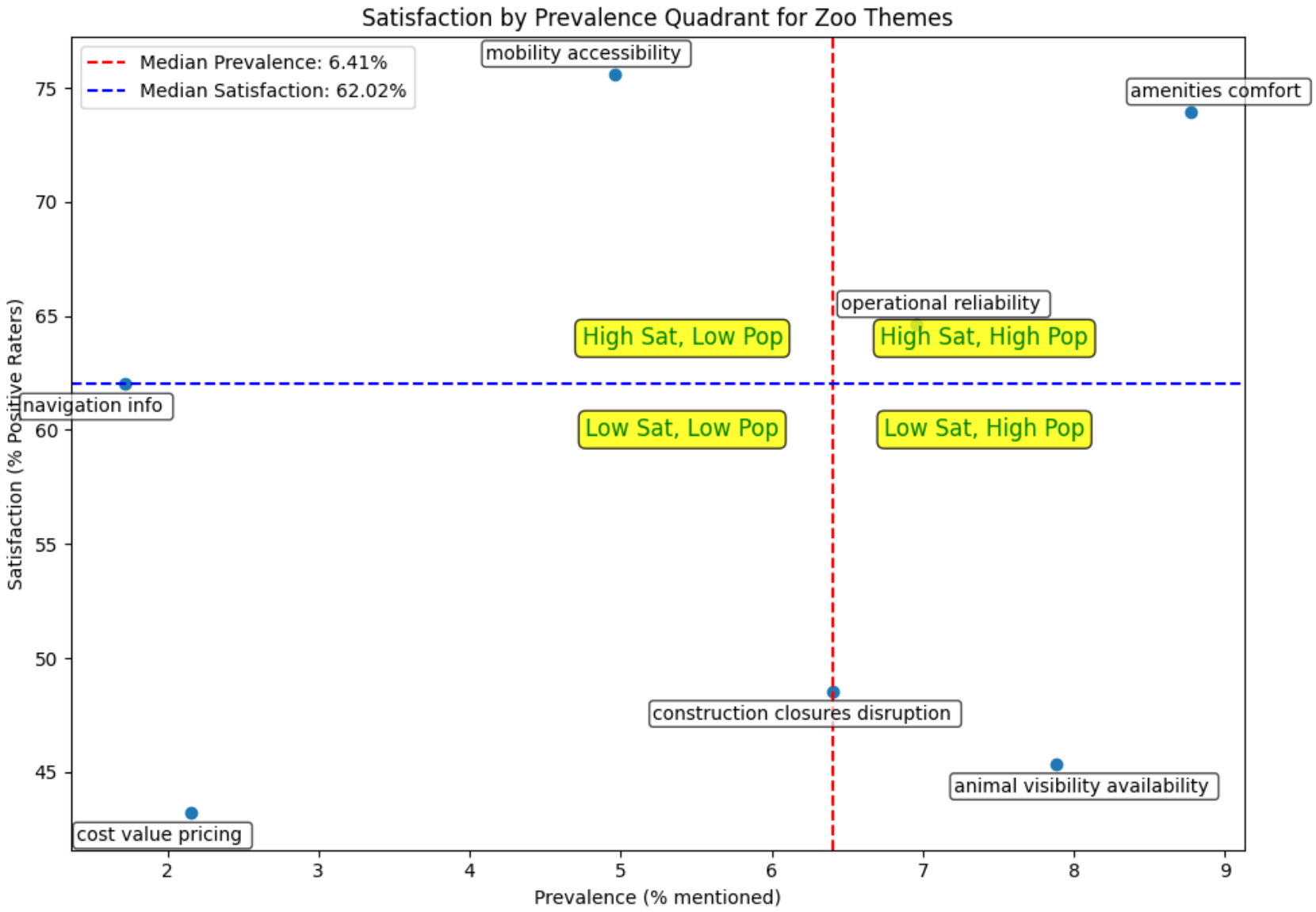
- An average satisfaction vs prevalence matrix highlights highlights possible pain points to prioritize among complaint themes.
- The X-axis is the percentage of respondents who had a particular type of complaint
- The Y-axis is the average satisfaction of customers who brought up a given complaint.
- The lower right quadrant shows complaints that are highly common and shared by the most dissatisfied customers.



Pain Points: Complaint Themes

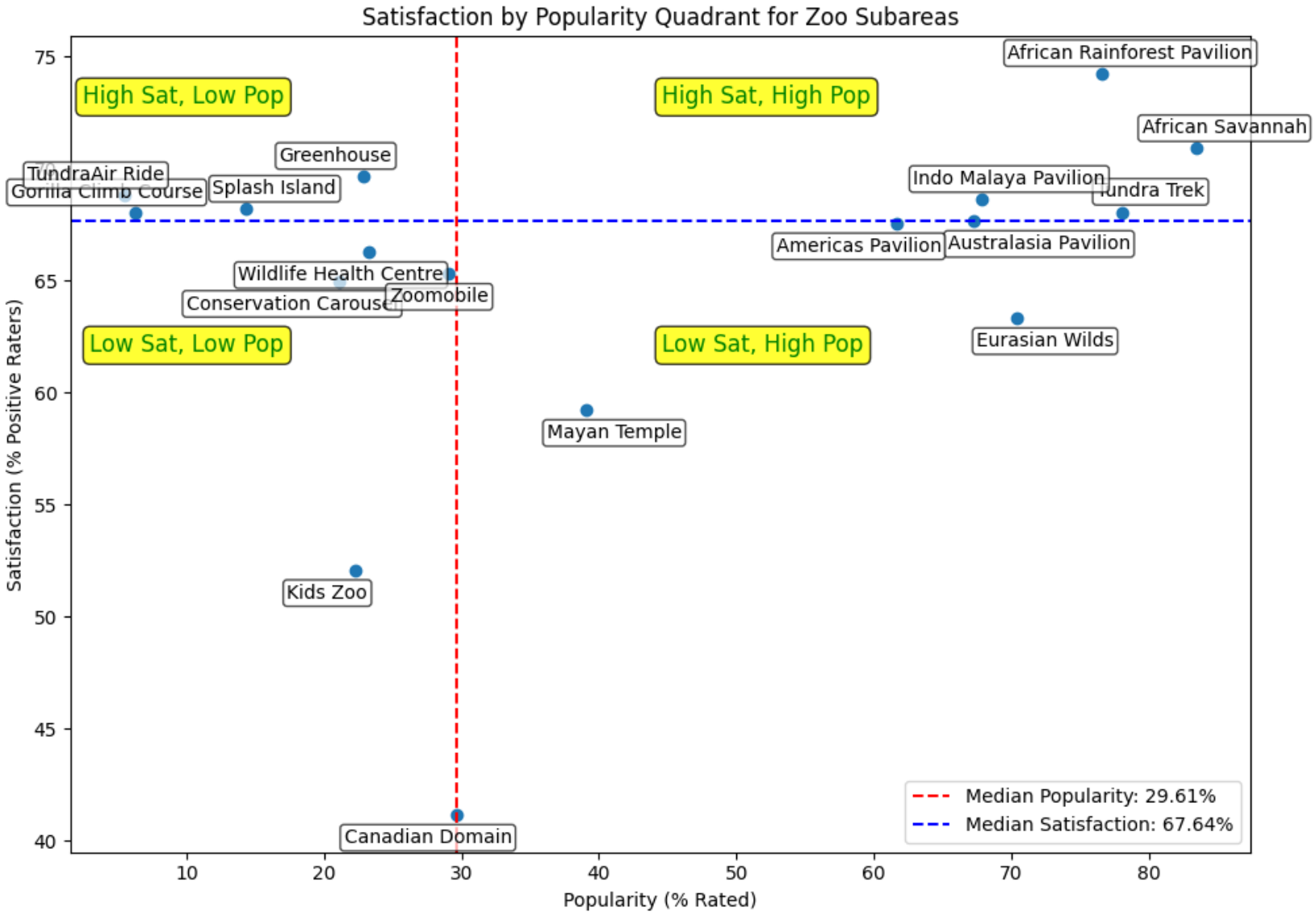
Notable Pain Points (lower right quadrant):

1. Animal Availability and Visibility
2. Construction Closures



Pain Points: Zoo Sub-Areas

- Similarly, a satisfaction by popularity matrix can highlight pain points and hidden gems among zoo subareas.
- The X-axis is the percentage of respondents attended a given subarea.
- The Y-axis is reported satisfaction respondants had for that specific subarea.
- The lower right quadrant shows well attended, but disliked sub areas (pain points
- the top left shows unattended but liked subareas (hidden gems)



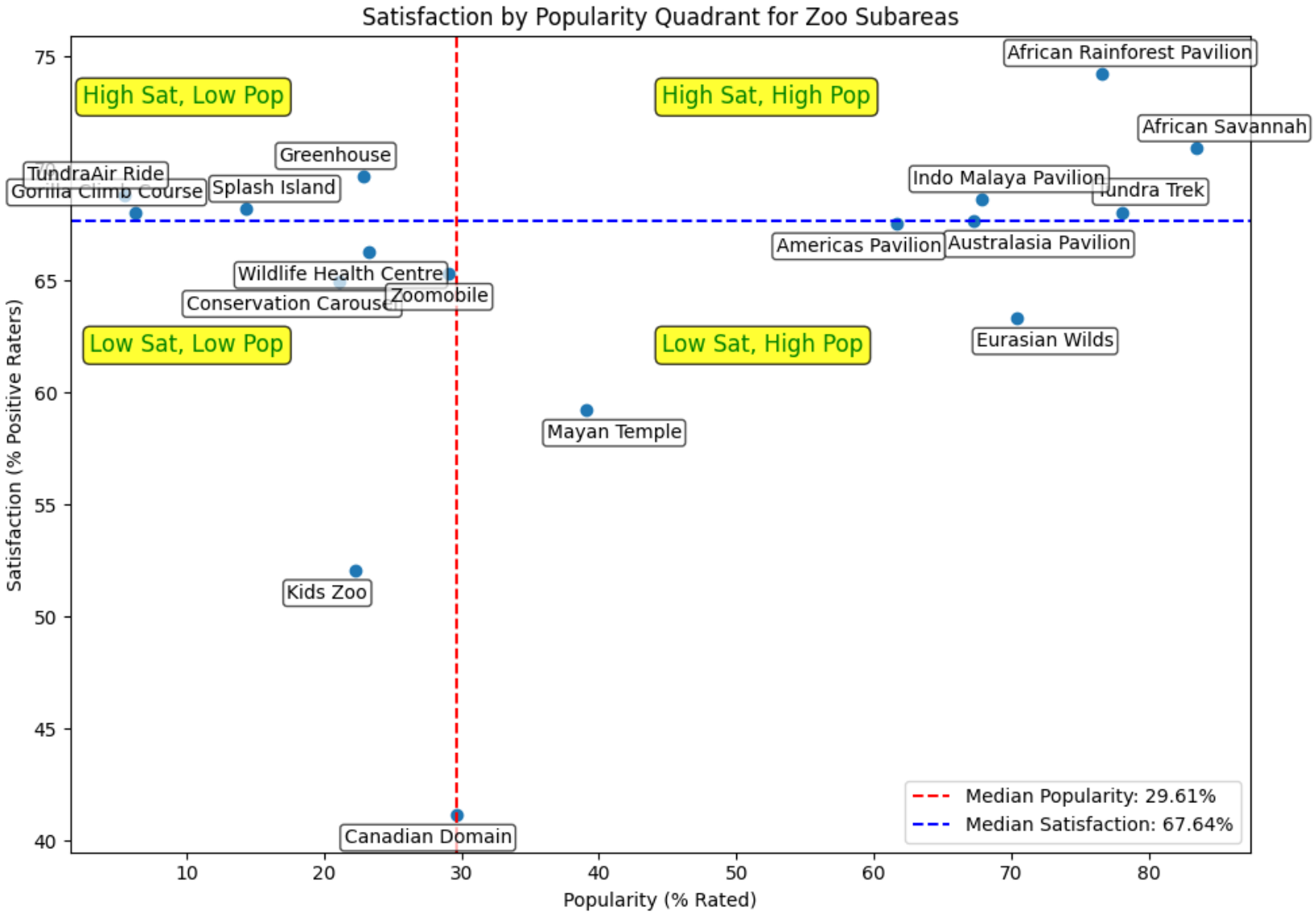
Pain Points: Zoo Sub-Areas

Pain Points (lower right quadrant):

1. Eurasian Wilds
2. Mayan Temple
3. Canadian Domain

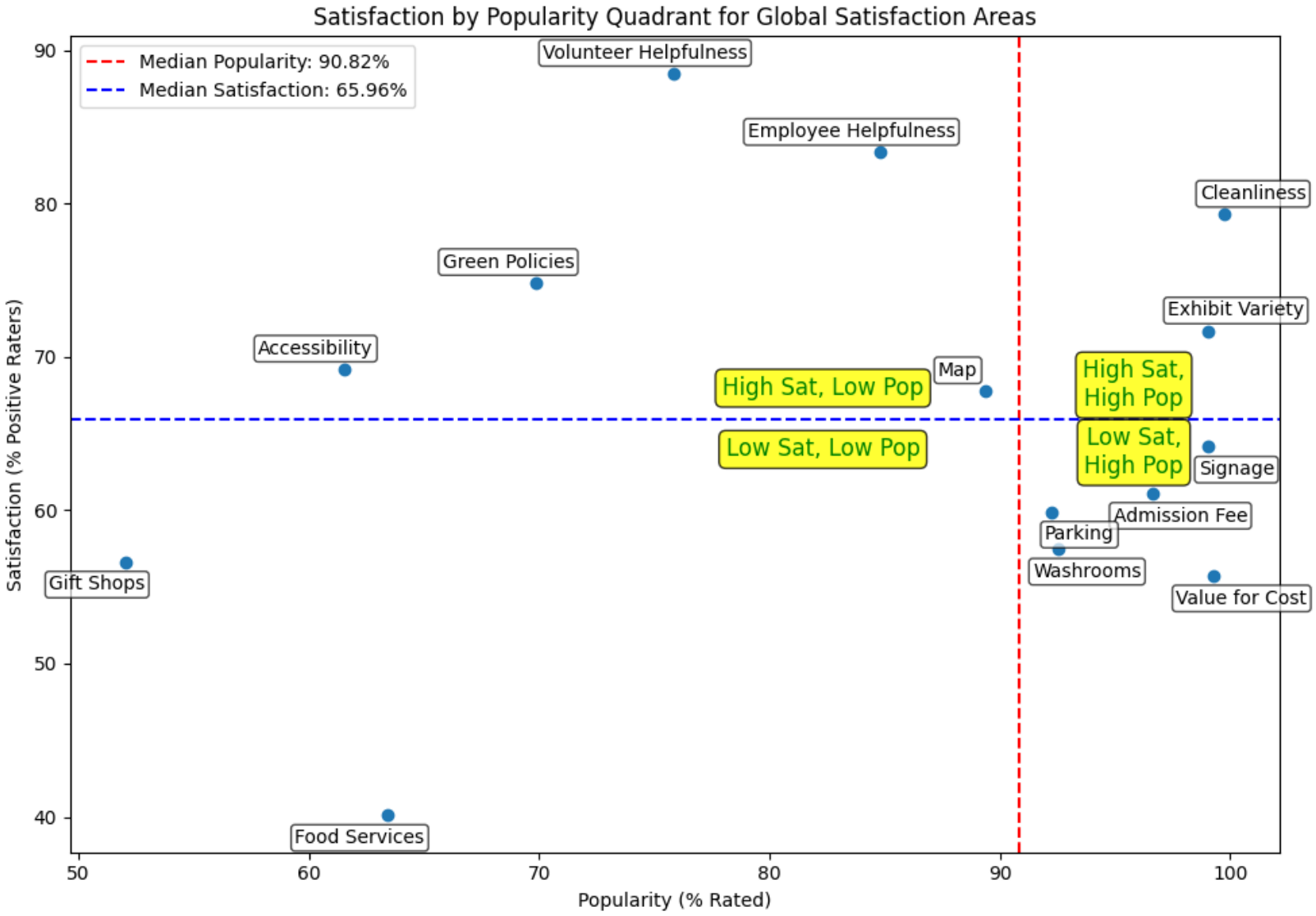
Hidden Gems (upper left quadrant):

1. TundraAir Ride
2. Splash Island
3. Greenhouse



Pain Points: Global Satisfaction Areas

- Finally, I highlighted pain points and hidden gems among sub-features of global satisfaction.
- The X-axis is the percentage of people who rated a given sub-feature, as a proxy for whether users experienced that areas.
- The Y-axis is the same as before.
- The bottom right quadrant shows near universal pain points of the zoo experience
- While the top left shows experience-specific hidden gems.



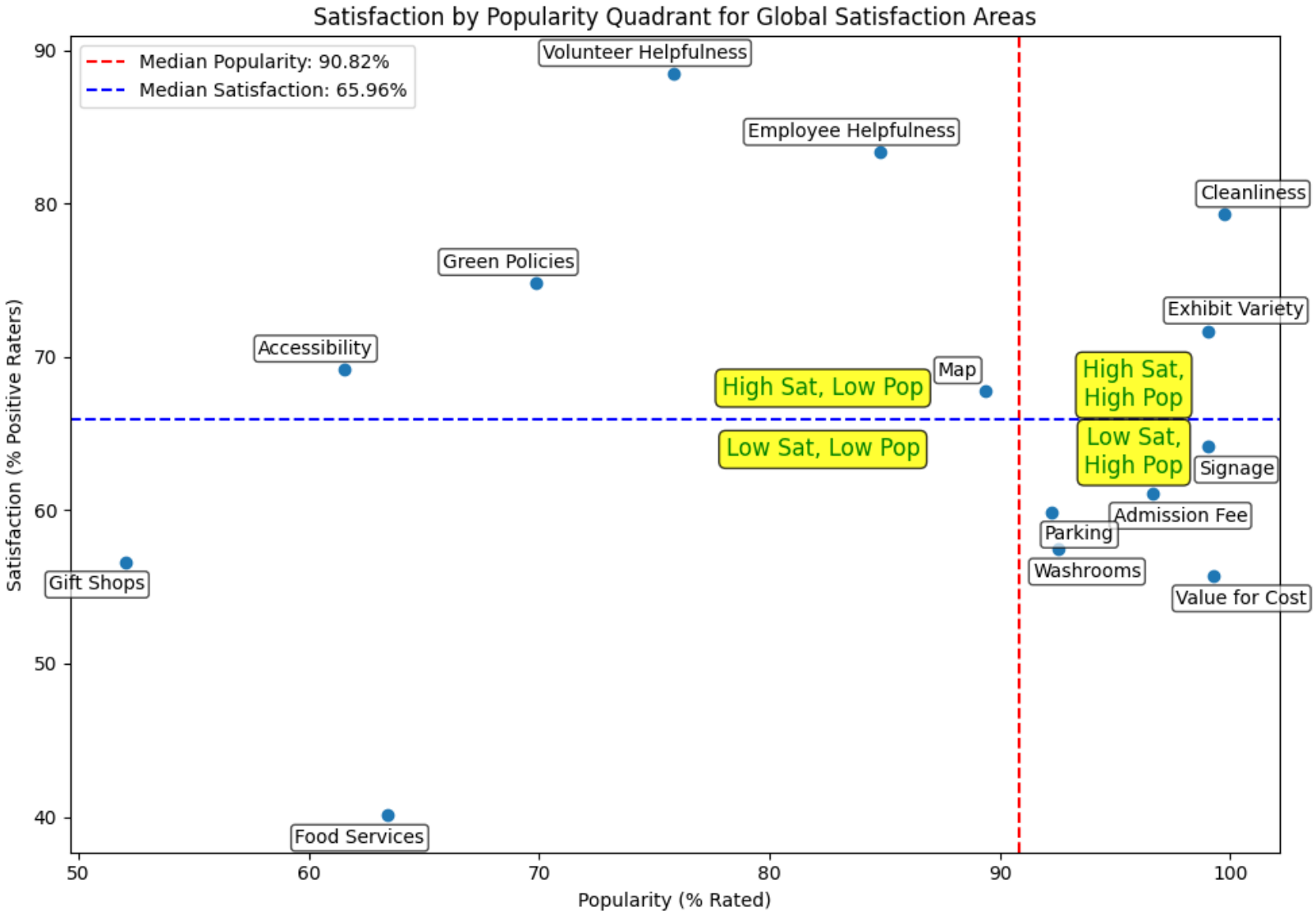
Pain Points: Global Satisfaction Areas

Pain Points (lower right quadrant):

1. Value for Cost
2. Washrooms
3. Parking
4. Signage

Hidden Gems (upper left quadrant):

1. Volunteer/Employee Helpfulness
2. Green Policies
3. Accessibility Policies



Pain Points: Summary

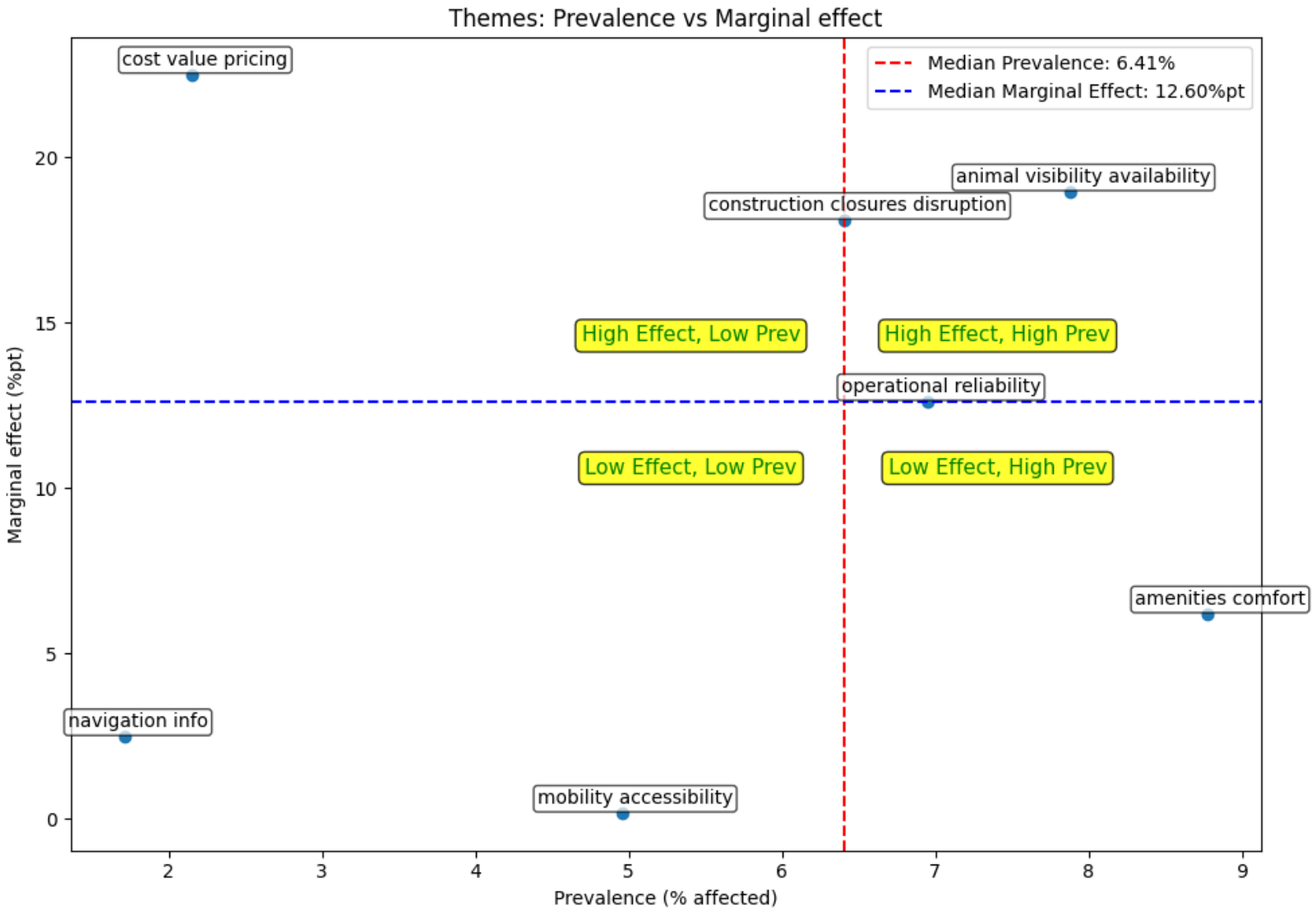
Domain	Pain Points	Hidden Gems
Complaint Themes	Animal Availability and Visibility, Construction Closures	-
Zoo Sub-Area	Eurasian Wilds, Mayan Temple, Canadian Domain	TundraAir Ride, Splash Island, Greenhouse,
Global Satisfaction	Value for Cost, Washrooms, Parking, Signage	Volunteer/Employee Helpfulness, Green Policies, Accessibility Policies

- Customers highlight pain points and hidden gems.
- However, we do not know the importance of these concerns.
- It is unclear how much increasing satisfaction in any domain will improve overall likelihood to recommend (or return to) the zoo.
- Key driver analysis will be conducted to translate pain points and hidden gems into areas of improvement.

Pinpointing Areas of Improvement

Areas of Improvement: Complaint Themes

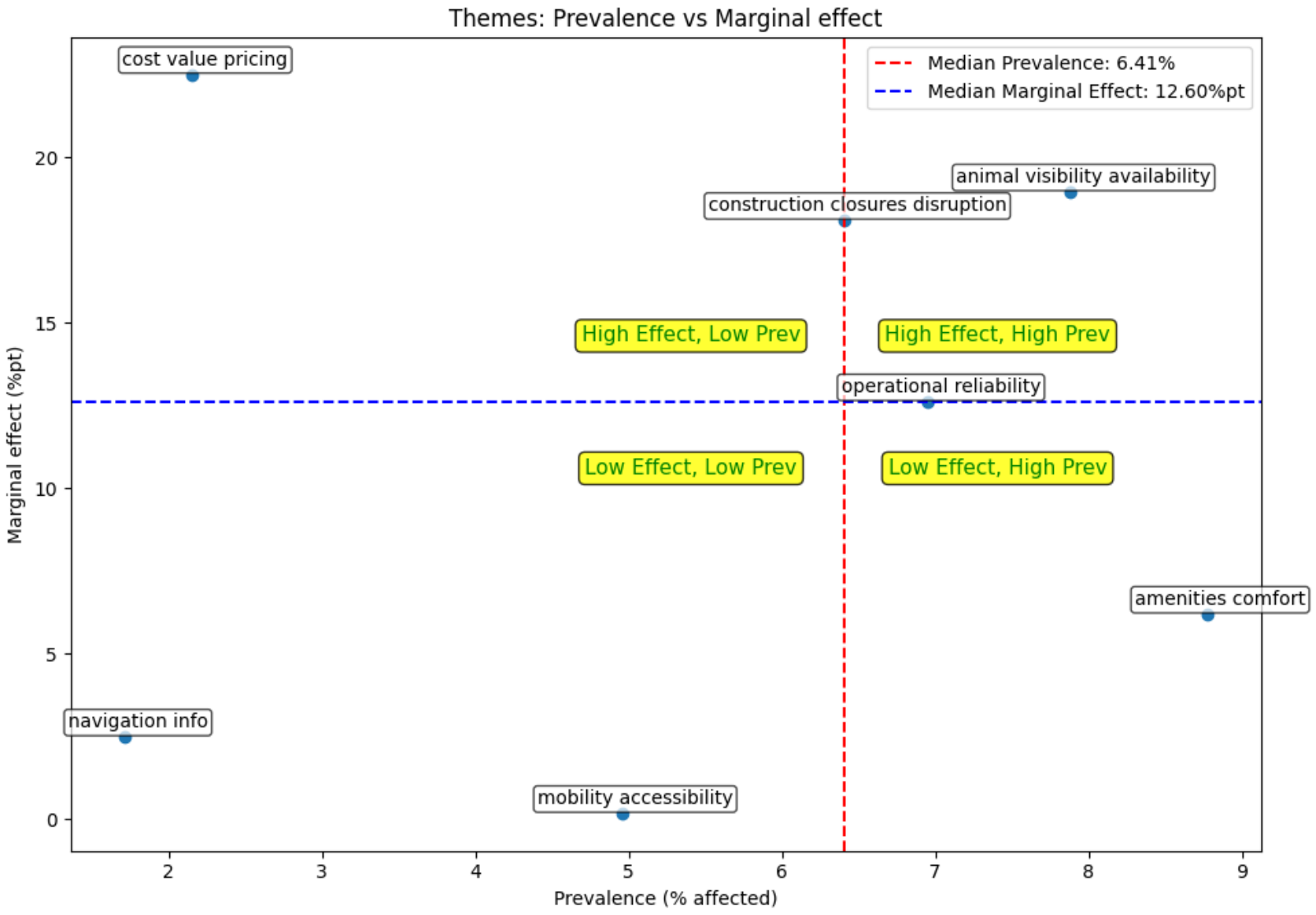
- I estimated how much more affected visitors would be likely to recommend the zoo if their complaints were addressed.
- The X-axis is the percentage of respondents who had a particular type of complaint
- The Y-axis is how much more those respondents would be likely to recommend the zoo if their complaints were addressed.
- The top right quadrant shows complaints that are highly common and that would lead to the largest % increase in zoo recommendations if addressed.



Areas of Improvement: Complaint Themes

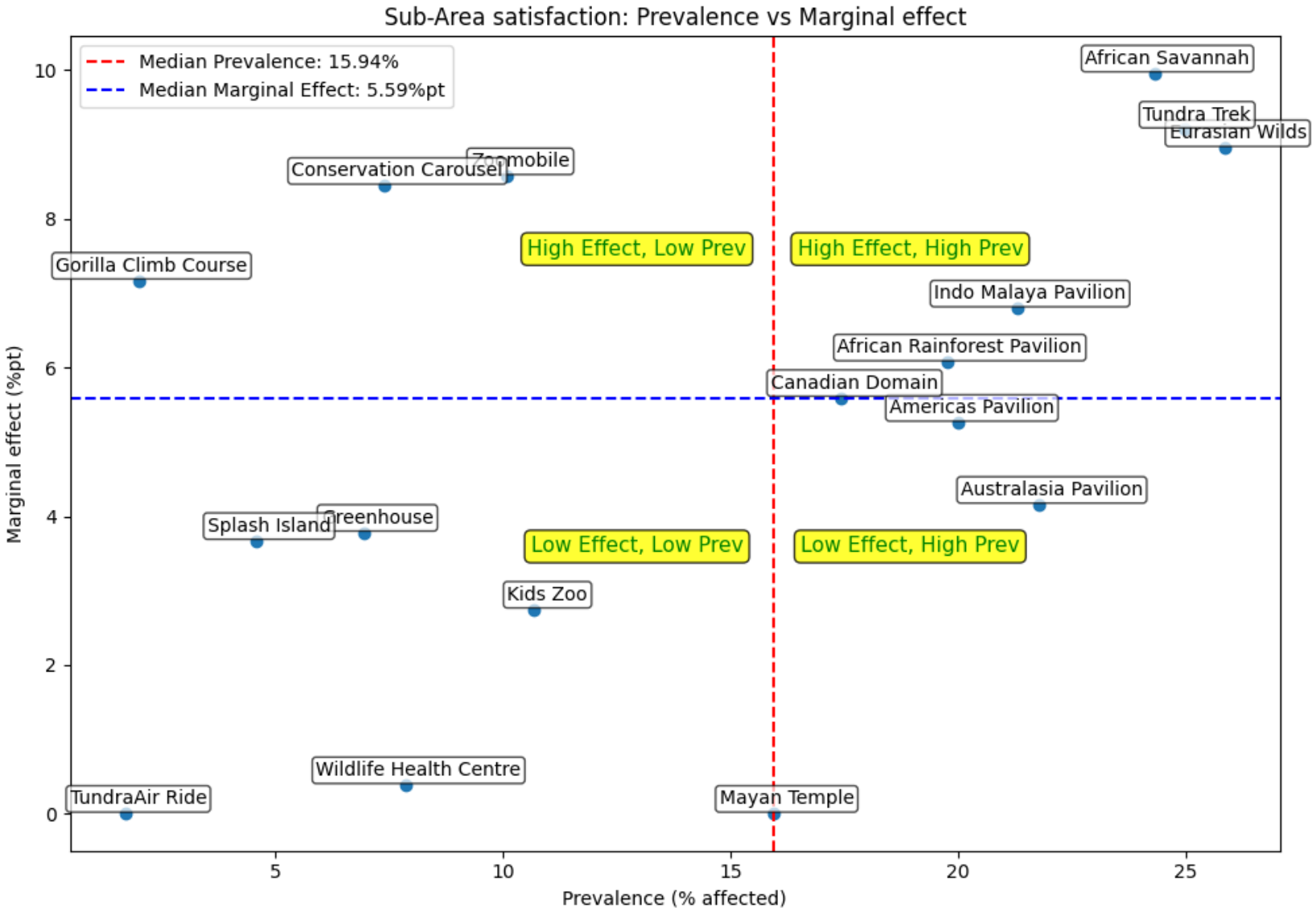
Top 3 Areas of Improvement (top right quadrant):

1. Animal Availability and Visibility
   - (est. 1.49% lift in total recommendations if addressed.)
2. Construction Closures
   - (est. 1.16% lift in total recommendations if addressed.)
3. Operational Reliability
   - (est. .88% lift in total recommendations if addressed.)



Areas of Improvement: Zoo Subareas

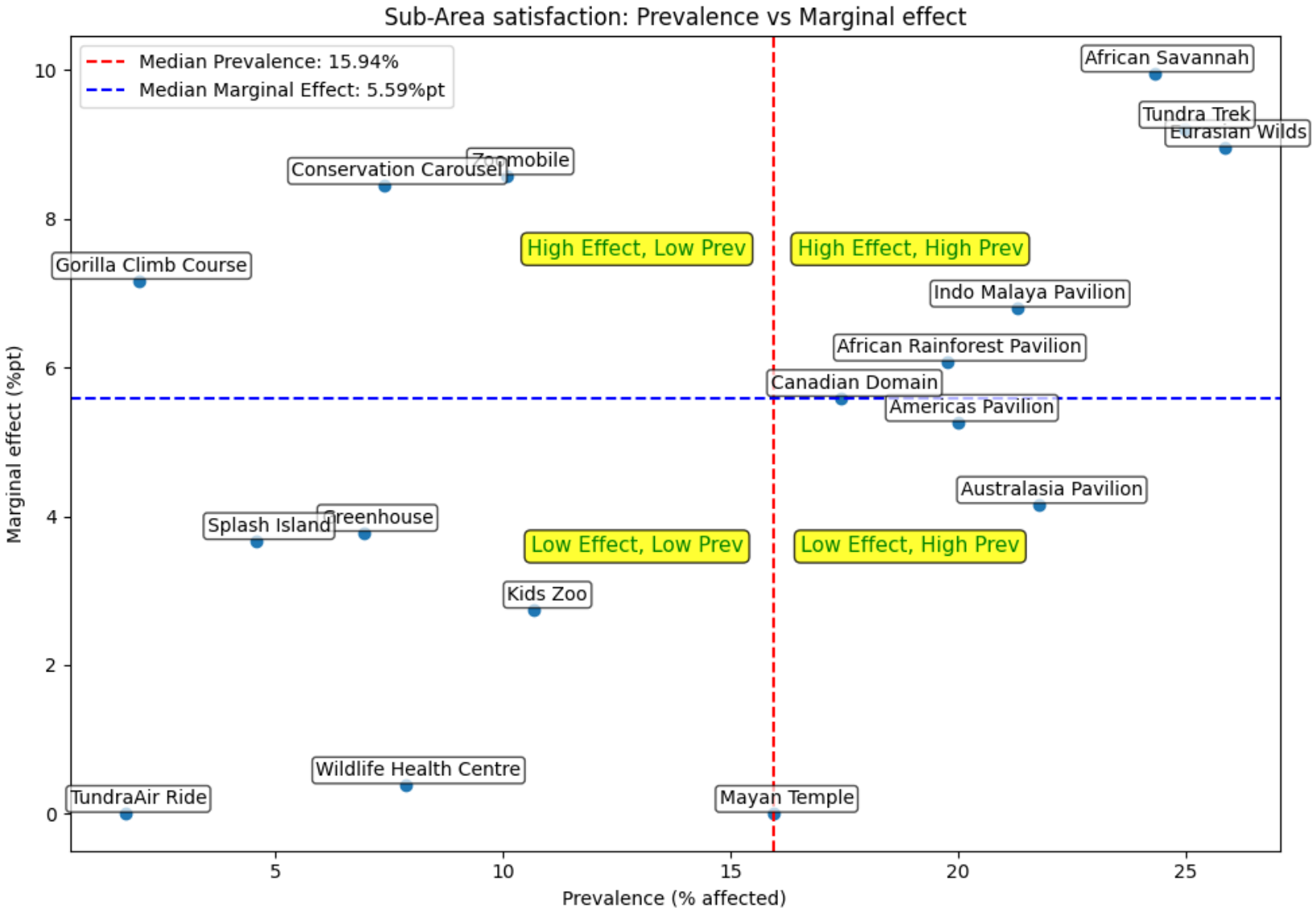
- The same analysis was repeated to see how much every point increase in satisfaction with a given zoo subarea would increase recommendation likelihood.
- The X-axis represents the percentage of survey respondents who reported by dissatisfied with a given subarea
- The Y-axis represents how much more likely those respondents would be to recommend the zoo if their satisfaction with the zoo increase by 1 point (out of 5).



Areas of Improvement: Zoo Subareas

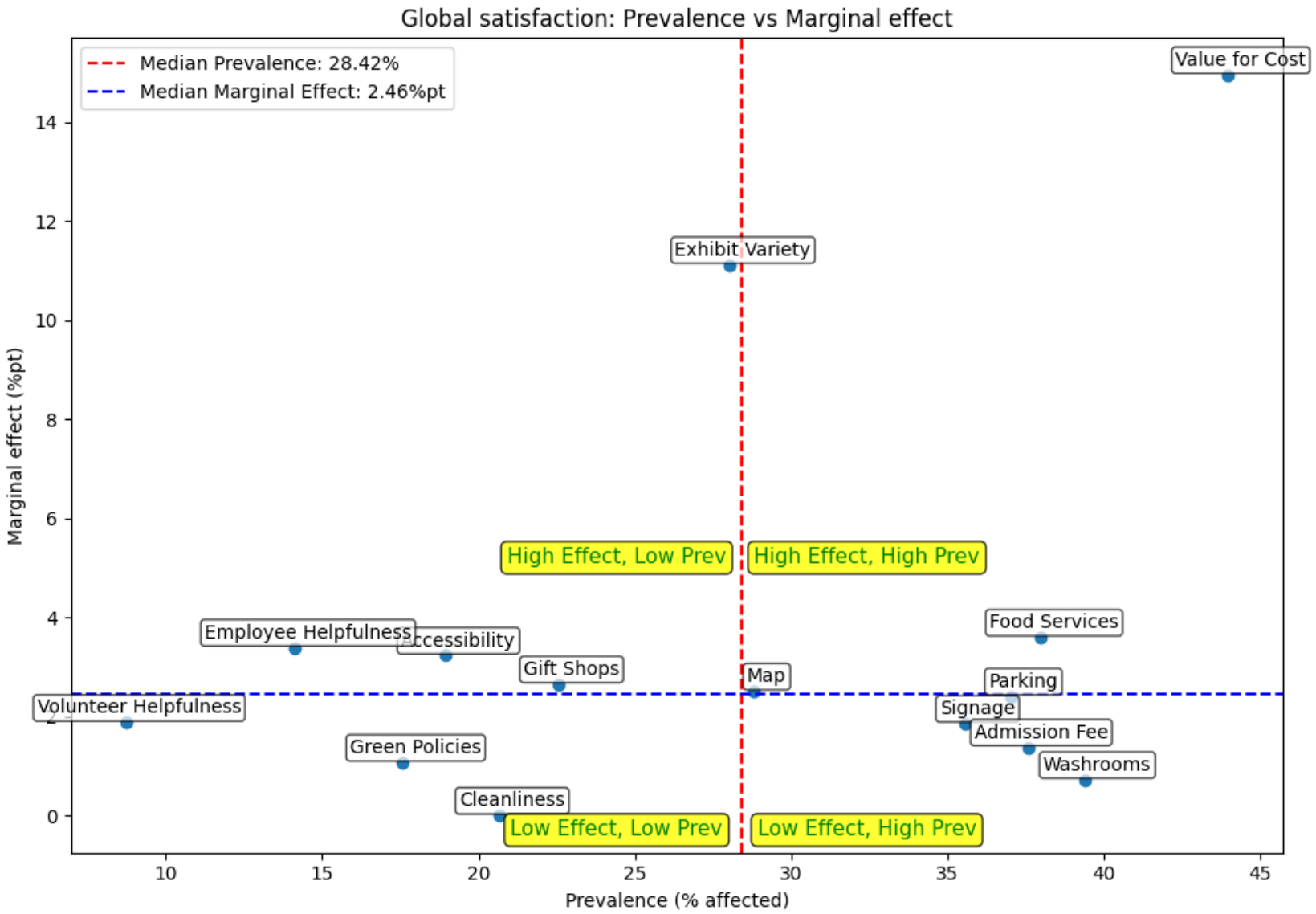
Top 3 Areas of Improvement (top right quadrant):

1. African Savannah
   - (est. 2.42% lift in total recommendations per 1 point increase in satisfaction.)
2. Eurasian Wilds
   - (est. 2.31% lift in total recommendations per 1 point increase in satisfaction.)
3. Tundra Trek
   - (est. 2.30% lift in total recommendations per 1 point increase in satisfaction.)



Areas of Improvement: Global Satisfaction

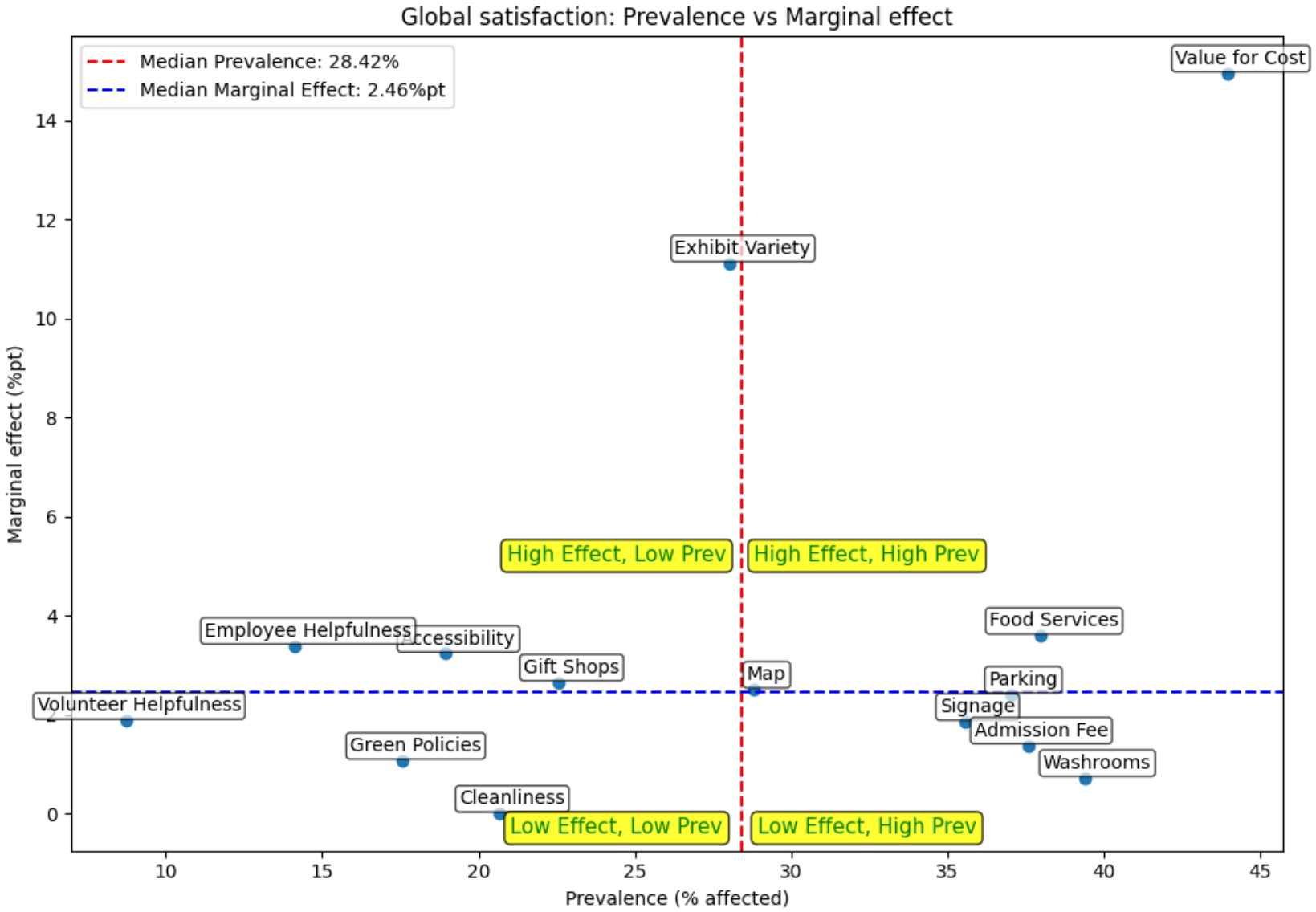
- Similarly, I examined which sub-features of global satisfaction with the zoo could be the most impactful if addressed.
- I estimated how much every point increase in satisfaction with a given sub-feature would increase recommendation likelihood.
- The X and Y axes are marked the same as before.



Areas of Improvement: Global Satisfaction

Top 3 Areas of Improvement (top right quadrant):

1. Value for Cost
   - (est. 6.57% lift in total recommendations per 1 point increase in satisfaction.)
2. Exhibit Variety
   - (est. 3.12% lift in total recommendations per 1 point increase in satisfaction.)
3. Food Services
   - (est. 1.37% lift in total recommendations per 1 point increase in satisfaction.)



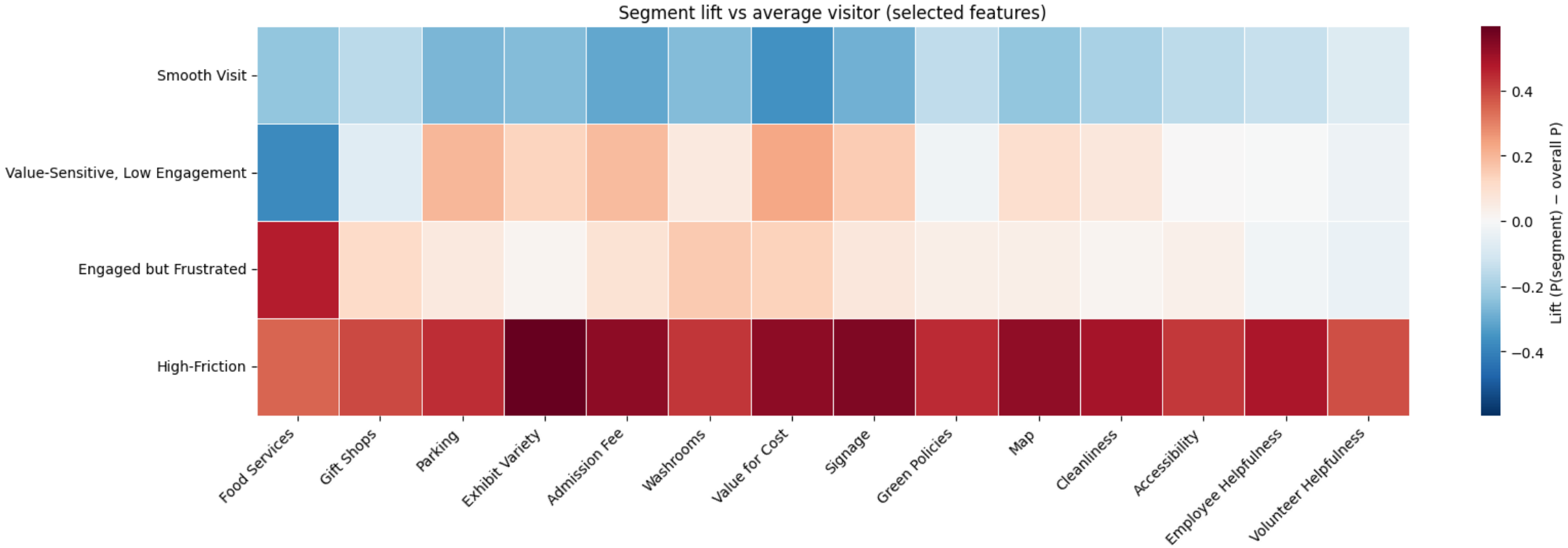
Areas of Improvement: Summary

Domain	Areas of Improvement
Complaint Themes	Animal Availability and Visibility, Construction Closures, Operational Reliability
Zoo Sub-Area	African Savannah, Eurasian Wilds, Tundra Trek
Global Satisfaction	Value for Cost, Exhibit Variety, Food Services

- Targeting the following areas of improvement would have the greatest increase in overall zoo recommendations (and perhaps visits).
- However, customer concerns will depend on individual experiences at the zoo.
  - For example, food service satisfaction would be more important to customers who plan to each lunch at the zoo.
- Thus, I will further segment customers into experience-specific personas.

Segmenting Customers into Distinct Personas

Customer Segmentation



4 distinct customer personas were determined using Latent Class Analysis:

1. Smooth Visit customers were globally satisfied across all domains
2. Value-Sensitive, Low Engagement customers wanted better value for cost, exhibit variety and parking experiences.
3. Engaged but frustrated customers were particularly dissatisfied with food services, gift shops and washrooms.
4. High Friction customers were dissatisfied with all aspects of their zoo experience

Customer Segmentation

Persona	Total Number	Percentage of Sample	Percent Members	Percent Local	Avg. Subareas Skipped	Percent Recommended
Smooth Visit	3094	41.14%	51.26%	70.24%	5.62	94.39%
Value-Sensitive, Low Engagement	1454	19.34%	43.48%	69.91%	6.00	50.63%
Engaged but Frustrated	2010	26.72%	37.06%	61.97%	5.35	58.75%
High-Friction	963	12.80%	23.42%	58.57%	4.08	13.79%

- The majority of the sample is made up by smooth-visit customers and only a small minority represent high-friction customers.
- However, a significant portion of customers were either value-sensitive, low engagement or engaged but frustrated by zoo amenities.
- Addressing the respective concerns of one or both of these personas could increase zoo recommendations.
- Value-sensitive, low engagement concerns (Exhibit Variety, Value for Cost) may be harder to implement.
- Therefore, engaged, but frustrated customers might be the easiest to address by improving amenity quality (Food Services, Washrooms, Gift Shops).